WEEK #1

1# Prepare a report on SageMath tool and its different functionalities.

SageMath (System for Algebra and Geometry Experimentation) is a free, open-source mathematics software system designed to support teaching, learning, and research. It integrates many powerful math libraries into a single, unified platform and uses Python as its main programming language.

Key functionalities

1. **Symbolic Mathematics**
   - o Algebraic simplification, factorization, expansion
   - o Solving equations and inequalities
   - o Calculus operations like differentiation and integration
2. **Algebra & Linear Algebra**
   - o Operations on matrices, vectors, and polynomials
3. **Graphing & Visualization**
   - o 2D and 3D plots
   - o Graphs of functions, curves, and surfaces
4. **Statistics & Probability**
   - o Descriptive statistics
   - o Probability distributions
5. **Programming Support**
   - o Based on Python
   - o Easy integration with other Python libraries
6. **Educational & Research Use**
   - o Useful for students, teachers, and researchers
   - o Supports mathematics, physics, computer science, and engineering

2# Write a Python program to detect whether given integer is perfect or not ?

```
n = int(input("enter num: "))

sum = 0

if n <= 1:

  print("Invalid input")

else:

  for i in range(1, n):

    if n % i == 0:

      sum =sum+i

  if sum == n:

    print("Perfect")
```

```
else:
print("Not Perfect")
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 1/q2.py"
enter num: 28
Perfect
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

3# Write a Python program to detect whether given integer is Armstrong Integer or not?

```
n = int(input("enter num: "))
if n <= 0:
    print("invalid input")
else:
    temp = n
    sum = 0
    digits = len(str(n))
    while temp > 0:
        digit = temp % 10
        sum = sum + digit ** digits
        temp = temp // 10
    if sum == n:
        print("armstrong")
    else:
        print("not")
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 1/q3.py"
enter num: 153
armstrong
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

WEEK #2

1# Write a program to detect whether a number is prime or not.

```
n = int(input("enter num: "))
if n <= 1:
   print("invalid input")
else:
   for i in range(2, n):
      if n % i == 0:
         print("not a prime number")
         break
   else:
      print("prime number")
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 2/q1.py"
enter num: 7
prime number
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

2# Write a function to compute GCD of two integers.

```
a = int(input("enter a: "))
b = int(input("enter b: "))
if a <= 0 or b <= 0:
   print("invalid input")
else:
   while b != 0:
      a, b = b, a % b
   print("GCD of a and b is:", a)
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 2/q2.py"
enter a: 21
enter b: 28
GCD of a and b is: 7
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

3# Write a program to detect whether two numbers are relatively prime or not?

```
a = int(input("enter a: "))
b = int(input("enter b: "))
if a <= 0 or b <= 0:
  print("invalid input")
else:
  while b != 0:
    a, b = b, a % b
  print("GCD of a and b is:", a)
  if a==1:
    print("numbers are relatively prime")
  else:
    print("numbers are relatively not prime")
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 2/q3.py"
enter a: 5
enter b: 7
GCD of a and b is: 1
numbers are relatively prime
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

4# Write a function to display all‘n’ narcissistic number.

```
def display_num(n):
  a = 10 ** (n - 1)
  b = 10 ** n
  for i in range(a, b):
    temp = i
    sum = 0
    while temp > 0:
      digit = temp % 10
```

```
        sum = sum + digit ** n
        temp = temp // 10
    if sum == i:
        print(i)
n = int(input("Enter value of n: "))
display_num(n)
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 2/q4.py"
Enter value of n: 3
153
370
371
407
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

WEEK #3

1# Write a function to check whether a number is in the form of 2k or not.

```
def check_num():
  n=int(input("enter number:"))
  if n<1:
    print("invalid")
  else:
    for k in range(0,n+1):
      if 2**k==n:
        print("it is in the form of 2**k")
        break
    else:
      print("it is not in the form of 2**k")
check_num()
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 3/q1.py"
enter number:128
it is in the form of 2**k
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

2# Write a function to check whether a prime number is Mersenne prime or not?

```
def check_mersenne():
  num = int(input("enter number: "))
  if num <= 1:
    print("invalid input")
    return
 for i in range(2, num):
  if num % i == 0:
      print("not a mersenne prime")
      return
  found = False
   for p in range(1, num):
    if 2**p - 1 == num:
      found = True
      break
   if not found:
    print("not a mersenne prime")
    return
  for i in range(2, p):
    if p % i == 0:
      print("not a mersenne prime")
      return
     print("mersenne prime")
check_mersenne()
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 3/q2.py"
enter number: 7
mersenne prime
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

3# Write a program to evaluate Euler Totients (Totatives) of a given integer.

```
import math
def Euler_Totients():
   n=int(input("enter number:"))
   if n<1:
      print("invalid input")
      return
   count=0
 for i in range(1,n):
      if math.gcd(i,n)==1:
         count=count+1
   print("euler totients of" , n , "is:" ,count)
Euler_Totients()
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 3/q3.py"
enter number:10
euler totients of 10 is: 4
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

4# Implement a function that takes two arguments (r, n) and computer order of r under modulo(n) operation.

```
import math
def modulo_operation(r, n):
   if r <= 0 or n <= 0:
      print("invalid input")
```

```
        return
    if math.gcd(r, n) != 1:
        print("order does not exist")
        return
    i = 1
    while (r**i) % n != 1:
        i += 1
    print("Order =", i)
r = int(input("enter r: "))
n = int(input("enter n: "))
modulo_operation(r, n)
```

```
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 3/q4.py"
enter r: 2
enter n: 5
Order = 4
PS C:\Users\ACER\Desktop\AHF\elective lab\WEEK 3>
```

WEEK #4

1# Illustrate major potential security vulnerabilities in web applications. Use a checklist to document the vulnerabilities and suggest basic countermeasures.

✓ **SQL Injection** When a website takes user input (like login form, search box) and directly inserts it into a database query without checking it properly, an attacker can insert harmful SQL commands

Countermeasure: Prepared statements → The SQL structure (the command) should be fixed. User input should be inserted safely — not mixed directly into the query.

Input validation → Reject special characters if not required.

✓ **Cross-Site Scripting (XSS)** Attacker injects JavaScript into a website. When users visit that page, the script runs in their browser

Countermeasure: Escape output → If a user types special characters like , ", ' We change them into a safe version so the browser does NOT treat them as code.

CSP (Content Security Policy) → Restrict which scripts can run.

✓ **Cross-Site Request Forgery (CSRF)** If you're logged into a site (like banking), attacker tricks your browser into sending a request without your knowledge

Countermeasure: Use CSRF tokens, A CSRF token is: A secret random value Generated by the website Sent with every important request (like form submission)

Use same-site cookies, A cookie is a small piece of data stored in your browser.It usually stores: Login session ID User preferences

✓ **Broken Authentication** Authentication system is weak.

Countermeasure: Strong passwords, MFA (Multi-Factor Authentication)

✓ **Broken Access Control** Users can access data they should not see.

Countermeasure: Role-based access control

✓ **Sensitive Data Exposure** Data like passwords, credit cards not properly protected.
Countermeasure: Use HTTPS, encrypt data, hash passwords.

2# Categorize the following hypothetical scenarios into different types of cybercrimes (e.g., hacking, identity theft, online fraud) and suggest how they can be prevented. a. Hacking b. Identity Theft c. Online Fraud d. Cyberstalking and Harassment e. Malware Attacks f. DDoS g. Eavesdropping h. Cyber Espionage i. Intellectual Property Theft j. Online Defamation k. Child Exploitation l. Cyberbullying

a. **Hacking**
   Type of Cybercrime: Unauthorized Access Crime Description: Illegal access to computer systems or networks without permission.
   Prevention: Use strong passwords, firewalls, regular software updates
b. **Identity Theft**
   Type of Cybercrime: Personal and Financial Data Crime Description: Stealing personal information to impersonate someone for illegal purposes.
   Prevention: Protect personal data, use strong authentication, avoid sharing sensitive information online.
c. **Online Fraud**
   Type of Cybercrime: Financial Cybercrime Description: Deceiving users online to steal money or financial details.
   Prevention: Verify websites, avoid suspicious links, enable transaction alerts and secure payment methods.
d. **Cyberstalking and Harassment**
   Type of Cybercrime: Personal Cybercrime Description: Repeated online harassment, threats, or monitoring of individuals.
   Prevention: Use privacy settings, block and report offenders, take legal action if required.
e. **Malware Attacks**
   Type of Cybercrime: System and Network Crime Description: Use of malicious software to damage or gain control of systems.

Prevention: Install antivirus software, avoid unknown downloads, keep systems updated.

f. **DDoS (Distributed Denial of Service)**
Type of Cybercrime: Network Disruption Crime Description: Flooding a server with traffic to make services unavailable.
Prevention: Use traffic filtering, load balancers, and DDoS protection services.

g. **Eavesdropping**
Type of Cybercrime: Data Interception Crime Description: Unauthorized interception of network communications.
Prevention: Use encryption (HTTPS, VPN), secure Wi-Fi networks.

h. **Cyber Espionage**
Type of Cybercrime: Corporate and National Cybercrime Description: Stealing confidential government or corporate information.
Prevention: Strong access control, network monitoring, employee awareness training.

i. **Intellectual Property Theft**
Type of Cybercrime: illegal and Economic Cybercrime Description: Illegal copying or distribution of software, designs, or creative work.
Prevention: Copyright protection, digital rights management.

j. **Online Defamation**
Type of Cybercrime: Reputation-Based Cybercrime Description: Posting false or harmful content to damage a person's reputation.
Prevention: reporting abusive content

k. **Child Exploitation**
Type of Cybercrime: Severe Cybercrime Description: Online abuse or exploitation of minors.
Prevention: Parental controls, awareness programs, strict laws, reporting mechanisms.

l. **Cyberbullying**
Type of Cybercrime: Social Cybercrime Description: Harassment or humiliation of individuals through digital platforms.
Prevention: Education, reporting tools, counseling, responsible online behavior.

3# Create a timeline of key events in the development of the Indian IT Act. Discuss any one section of the act (e.g., Section 46, 66, 66A) with its implications in combating cybercrime.

TIMELINE OF KEY EVENTS IN THE DEVELOPMENT OF THE INDIAN IT ACT

- 1996- The United Nations Commission on International Trade Law (UNCITRAL) adopted the Model Law on Electronic Commerce, which inspired India to frame cyber laws.
- 1998- The first draft of the Information Technology Bill was introduced in the Indian Parliament to regulate e-commerce and cyber activities.
- 2000- The Information Technology Act, 2000 was passed by the Indian Parliament and came into force on 17 October 2000. It gave legal recognition to electronic records and digital signatures.

- 2004- Cyber Appellate Tribunal was established to handle cyber law disputes and appeals. 2008 The Information Technology (Amendment) Act, 2008 was passed to address new forms of cybercrime such as identity theft, cyber terrorism, and online fraud. New sections like 66A, 66B, 66C, and 66D were added.
- 2009- The amended IT Act, 2008 came into force, strengthening provisions related to data protection and cybercrime.
- 2015- Section 66A of the IT Act was struck down by the Supreme Court of India in the Shreya Singhal case for violating freedom of speech.

SECTION 66 OF THE INFORMATION TECHNOLOGY ACT

• Section 66 deals with computer-related offences committed dishonestly or fraudulently.
• It includes unauthorized access, data theft, data alteration, spreading viruses, and damage to computer systems.
• Punishment includes imprisonment up to three years, fine up to five lakh rupees, or both.
• This section acts as a deterrent against hacking and misuse of computer systems.
• It helps protect individuals, organizations, and government data from cybercrime.

4# Draft a basic security policy for a small office setup, focusing on password policies, data backup policies, and internet usage guidelines.

BASIC SECURITY POLICY FOR A SMALL OFFICE SETUP

• PASSWORD POLICY
Use strong passwords with at least 8 characters. Include uppercase, lowercase, numbers, and symbols. Do not share passwords with anyone. Change passwords regularly (every 60–90 days).

• DATA BACKUP POLICY
Take regular backups of important data. Store backups on external drives or secure cloud storage. Keep at least one backup copy off-site. Protect backup data using passwords or encryption.

• INTERNET USAGE GUIDELINES
Use internet only for official work purposes. Avoid visiting unsafe or unauthorized websites. Do not download files or software from unknown sources. Do not open suspicious emails or links.

5# Write a program to test password strength based on length, complexity, characters, and patterns. Use it to evaluate common passwords.

```
def check_password(password):
    common_passwords = ["password", "123456", "12345678"]
    score = 0
    if len(password) >= 8:
        score += 1
    if len(password) >= 12:
        score += 1
    if any(ch.isupper() for ch in password):
        score += 1
    if any(ch.islower() for ch in password):
        score += 1
    if any(ch.isdigit() for ch in password):
        score += 1
    if any(not ch.isalnum() for ch in password):
        score += 1
    if password.lower() in common_passwords:
        return "Very Weak (Common Password)"
    if score <= 2:
        return "Weak"
    elif score <= 4:
        return "Medium"
    else:
        return "Strong"
password = input("enter your password: ")
result = check_password(password)
print("password strength:", result)
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\ACER> & C:/Users/ACER/AppData/Local/Programs/Python/Python310/python.exe "c:/Users/ACER/Desktop/AHF/elective lab/WEEK 4/w4q5.py"
enter your password: AsadHusain@123
password strength: Strong
PS C:\Users\ACER>
```

WEEK #5

1# Perform a basic SWOT analysis (Strengths, Weaknesses, Opportunities, Threats) for a sample organization's IT security setup.

SWOT Analysis of an Organization’s IT Security Setup

1. Strengths: The organization uses firewall, antivirus, and strong passwords to protect data. It also takes regular backups and has an IT team to manage security.

2. Weaknesses: Employees are not properly trained about cyber threats. Some software updates are delayed. Personal devices used by employees may not be fully secure.

3. Opportunities: The company can improve security by giving training to employees, using advanced security tools, and doing regular security testing.

4. Threats: The organization may face cyber-attacks like phishing, ransomware, hacking, and data breaches.

2# Write a program to simulate simple brute force attack. Try to crack numeric passwords of length 4,6 and 8 digits

```
password = input("Enter password: ")
if len(password) not in (4, 6, 8):
    print("Invalid password length")
elif not password.isdigit():
    print("Invalid password (must contain only digits)")
else:
    print("Starting brute force attack !")
    length = len(password)
    max_range = 10 ** length
    attempts = 0
    for i in range(max_range):
        attempts += 1
        guess = str(i).zfill(length)
        if guess == password:
            print("Password found:", guess)
            print("Total attempts:", attempts)
            Break
```

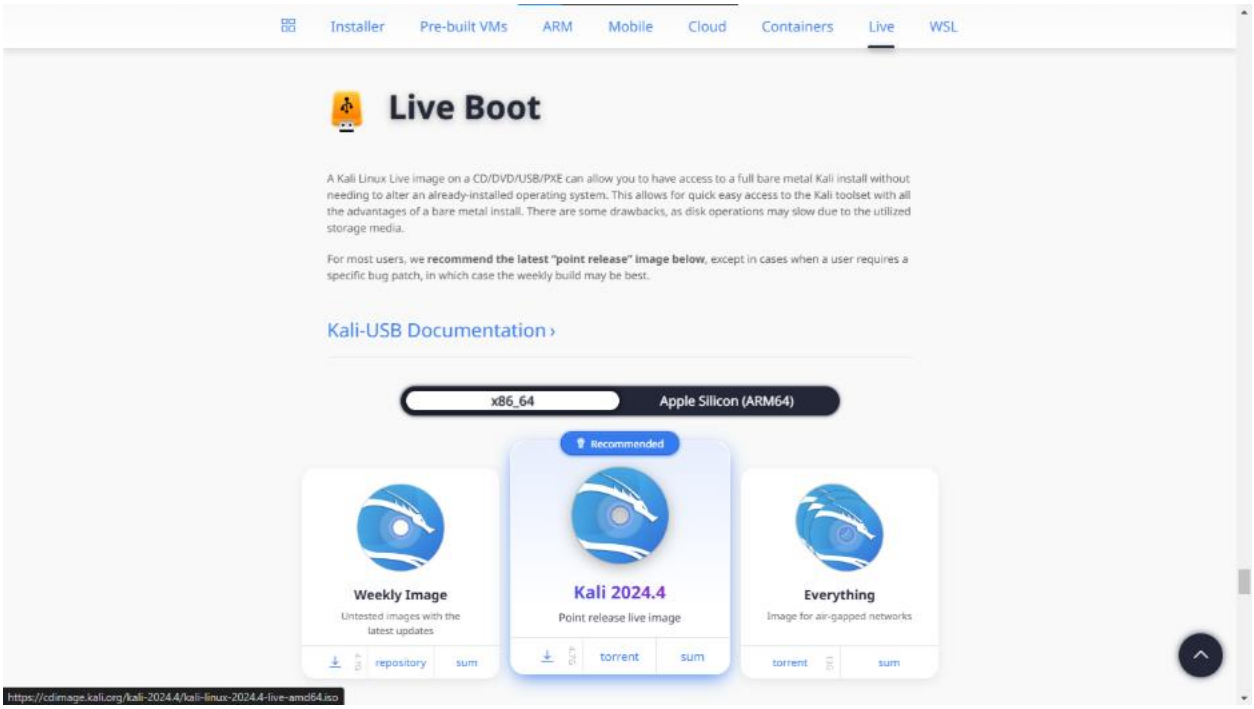
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Microsoft VS Code> & D:\python.exe "c:/Users/Asus/Desktop/college/6 SEM/elective lab/WEEK 5/q2.py"
Enter password: 9763
Starting brute force attack !
Password found: 9763
Total attempts: 9764
PS D:\Microsoft VS Code>
```

3# Install Kali Linux in your USB drive and demonstrate booting of the system using it. Write all the steps with proper screenshots.

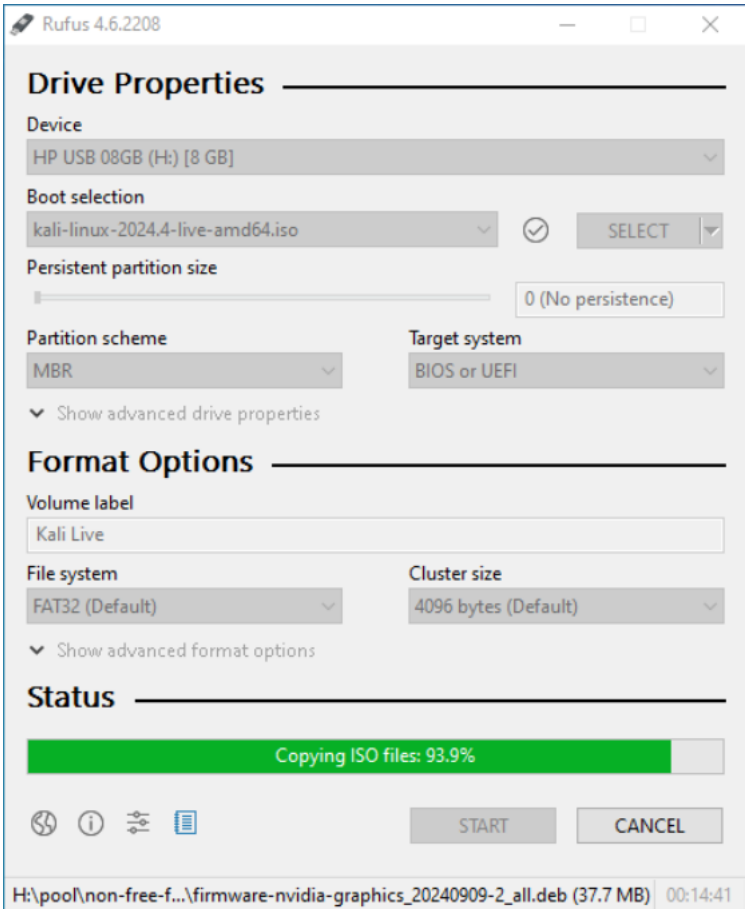
Step 01: Download Kali Linux

Download kali Linux Live Image (point release) from https://www.kali.org/get-kali/#kali-live



Step 02: Create a Bootable Kali Linux USB Drive

- Insert your USB drive.
- Open Rufus (can be downloaded from https://rufus.ie/en/)
- Select your USB under **Device**.
- Click on **SELECT** and choose the Kali Linux ISO.
- Set **Partition Scheme** to **MBR** (for legacy BIOS) or **GPT** (for UEFI).
- Keep the **File System** as **FAT32**.
- Click **START** and wait for the process to complete.



Step 03: Boot from USB

- **Restart your computer** and enter the **BIOS/UEFI** (Press **F2, F12, Esc, or Del**, depending on your system).
- Go to the **Boot Menu** and select the USB drive as the first boot device.
- Save and exit.
- Restart and select **Live Mode** or **Install Kali** to boot into Kali Linux.

4# Demonstrate the working of Wireshark tool to capture a sample TCP/IP packet details in Kali Linux. Write all steps with proper screenshots.

Step 01: Install Wireshark

Wireshark is pre-installed in Kali Linux, but if it's missing, install it using:

```
sudo apt update
sudo apt install wireshark -y
```

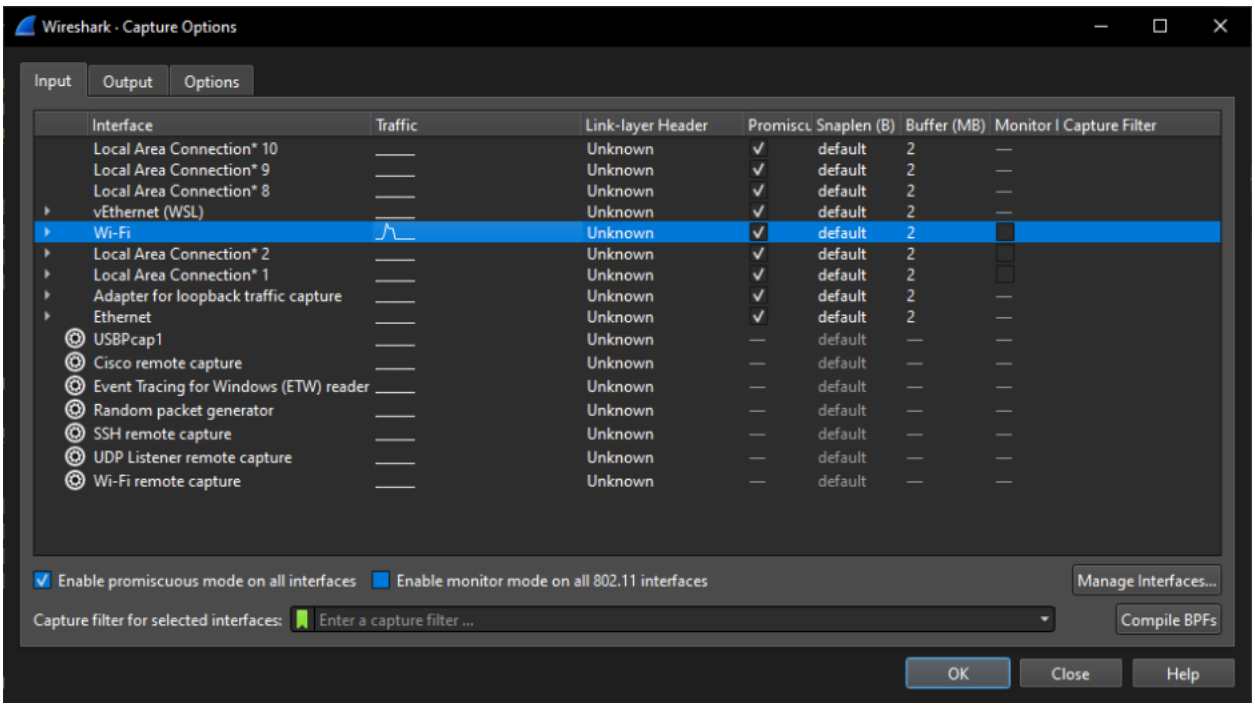
Step 02: Run Wireshark as Root/Sudo

To capture packets, you need administrative privileges. Start Wireshark using:

```
sudo wireshark
```

Step 03: Select Network Interface

- In the Wireshark window, go to **"Capture" → "Options"**.
- Select the appropriate **network interface**
- Click **OK** to begin packet capture.



Step 04: Generate a Sample TCP packet

To capture a **TCP/IP** packet, open a terminal and run:

```
curl http://google.com
```

or

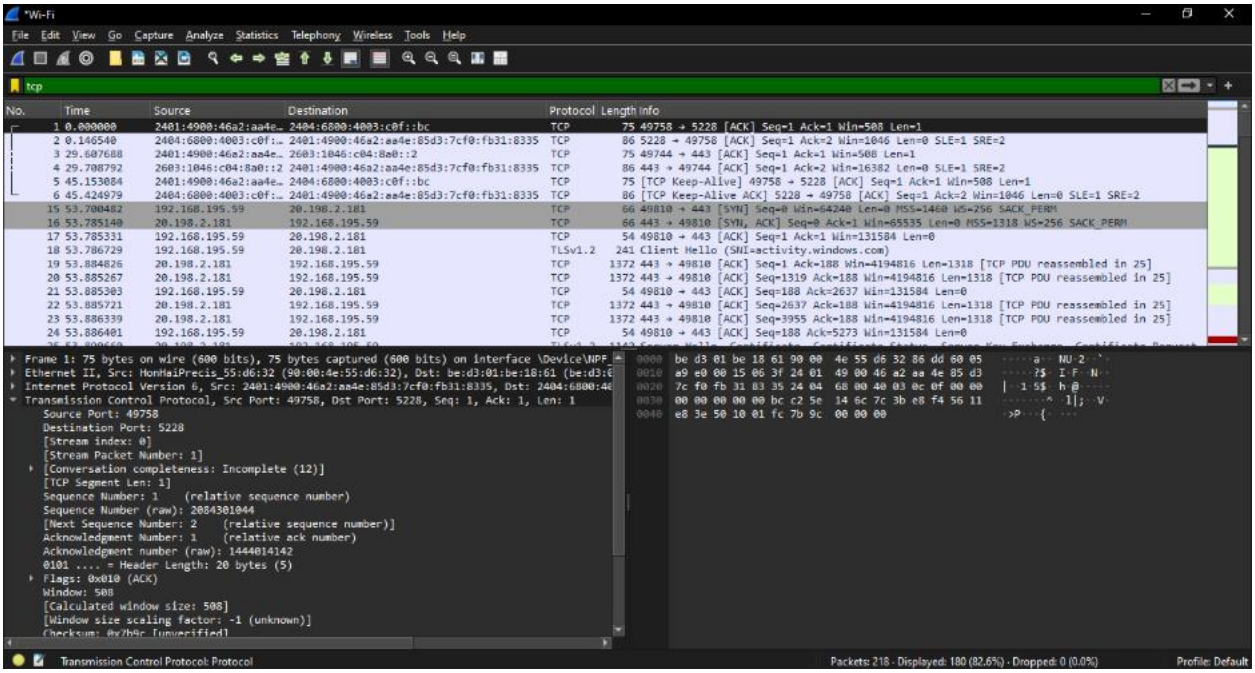
```
ping -c 4 google.com
```

This will generate network traffic, including TCP packets.

Step 05: Analyze Captured Packets

- Stop the capture by clicking the **Stop** button (red square).
- In the **Packet List pane**, look for **TCP packets**.
- Click on a **TCP packet** to view details.
- Expand the **Packet Details pane** to analyze:
  - **Ethernet Header** (Source & Destination MAC)
  - **IP Header** (Source & Destination IP, Protocol)
  - **TCP Header** (Ports, Flags, Sequence, Acknowledgment numbers)

Step 06: Filter and Export Packets

- Use filters like:
  - `Tcp`
  - `ip.addr == 192.168.1.1`
  - `tcp.port == 80`

- Save the capture using:
    - **File → Save As** (**.pcapng** format).